

ENHANCING HOME REHABILITATION FOR STROKE PATIENTS AND CEREBRAL PALSY THROUGH EMG CONTROLLED GAME BASED THERAPY

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ABSTRACT— Enhancing home rehabilitation for stroke patients and individuals with cerebral palsy through EMG-controlled game-based therapy is an essential step in the current situation. Stroke patients and individuals with cerebral palsy often face challenges in accessing regular rehabilitation services, leading to limited opportunities for recovery and improved motor function. It enables therapist to precisely monitor muscle activity and provide valuable insights for targeted interventions and personalized treatment plans. It offers real-time feedback and data-driven insights, it optimizes the effectiveness of therapy sessions, leading to faster recovery and improved functional abilities. It offers an immersive and interactive experience for patients. The ability to perceive their muscle activity and progress fosters motivation and engagement visually and audibly, resulting in increased compliance and better treatment results. It is an innovative approach that improves rehabilitation outcomes and paves the way for future advancements in the field. The proposed system's design and development are detailed, emphasizing user-friendly interfaces and adaptive game mechanics tailored to individual patient needs and progress. The outcomes of this study hold implications for healthcare practitioners, researchers, and policymakers, emphasizing the importance of incorporating innovative technology to improve rehabilitation outcomes and enhance the overall quality of life for patients with neurological disorders.

KEYWORDS: *Home Rehabilitation, EMG Controlled game, Monitor Muscle Activity, Neurological Disorder.*

I. INTRODUCTION

Cerebral palsy and stroke are two distinct neurological conditions that impact brain function and motor control. Cerebral palsy is a non-progressive disorder that typically manifests in early childhood, affecting muscle coordination and movement due to brain damage or abnormalities occurring during pregnancy, birth, or early development.[1] Symptoms vary widely, including muscle stiffness, weakness, and coordination difficulties. In contrast, stroke occurs when there is an interruption in the brain's blood supply, leading to cell damage.[2] It can affect individuals of any age and is often linked to risk factors such as high blood pressure, smoking, and diabetes. Stroke symptoms may include sudden numbness,

weakness, and difficulty speaking.[3] While these conditions are different, they share common challenges related to motor control, making early diagnosis and specialized rehabilitation essential for effective management.

The World Health Organization (WHO) recognizes cerebral palsy as a movement and posture disorder caused by early brain damage and emphasizes the importance of early intervention, rehabilitation, and inclusive support. Similarly,[4] WHO identifies stroke as a major global health concern, advocating for prevention, timely treatment, and rehabilitation programs to improve patient outcomes. Rehabilitation, particularly home-based programs, plays a crucial role in enhancing the quality of life for individuals affected by these conditions.[5] Home rehabilitation focuses on improving mobility, strength, and functional independence, allowing patients to regain essential skills for daily living.

A promising advancement in home rehabilitation is the integration of Electromyography (EMG)-controlled game-based therapy, which leverages EMG technology to measure muscle activity through electrodes.[7] This innovative approach enables interactive rehabilitation exercises tailored to individual needs. For stroke patients, EMG-controlled games help retrain specific muscle groups, enhancing motor function and promoting recovery. For individuals with cerebral palsy, the therapy can be customized to target muscle strength, coordination, and mobility.[8] By merging technology with therapy, this system not only increases motivation but also provides real-time data on muscle activity, allowing healthcare professionals to monitor progress and adjust treatment plans accordingly.[9] The flexibility of home-based EMG therapy ensures that patients can continue rehabilitation in a familiar environment, making the recovery process more engaging and effective. With its potential to enhance patient adherence, optimize rehabilitation outcomes, and promote faster recovery, EMG-controlled game-based therapy represents a significant step forward in neurological rehabilitation.

II. MATERIALS AND METHODS

A. HARDWARE COMPONENTS:

1) EMG Sensor:

Function: Captures and measures the electrical signals produced by muscle activity. These signals are used to control the game, providing real-time feedback and helping patients improve their motor skills.

2) Arduino:

Function: Acts as the central processing unit that receives signals from the EMG sensor, processes them, and translates them into game commands. The Arduino ensures that the system responds accurately and swiftly to the patient's muscle activity.

3) Battery:

Function: Powers the entire system, ensuring that the Arduino and EMG sensor operate continuously during therapy sessions. The battery must have sufficient capacity to support extended use, providing reliability for home-based rehabilitation.

4) Connecting Wires:

Function: Provide electrical connections between the EMG sensor, Arduino, and battery. They ensure stable and reliable communication and power distribution throughout the system, maintaining the integrity of the therapy setup.

B. SOFTWARE COMPONENTS:

1) Pygame

Pygame is a set of Python modules designed specifically for game development. It provides functionalities such as creating game loops, handling events, and rendering graphics and sound. Pygame is highly useful for beginners due to its simplicity and ease of use.

2) Dlib

Dlib is a powerful machine learning library written in C++, with a Python interface. It includes a wide range of machine learning algorithms, including tools for image processing and computer vision. Dlib is well-known for its robust face detection and face recognition capabilities.

3) Face_Recognition

The face_recognition library in Python is built on top of Dlib and provides a simple interface for facial recognition tasks. It can identify and manipulate faces with just a few lines of code, making it extremely user-friendly for developers working on face recognition projects.

4) CMake

CMake is a cross-platform tool that helps manage the build process of software projects. It simplifies the configuration and generation of build files for various

platforms. CMake is commonly used with C++ projects and is integral to building applications like OpenCV.

5) OpenCV_Python

Open-CV (Open Source Computer Vision Library) is a widely-used library for computer vision and image processing. The opencv-python package provides Python bindings for OpenCV, allowing developers to use its functionalities directly in Python. OpenCV includes algorithms for object detection, image segmentation, and more.

6) NumPy

NumPy is a fundamental library for scientific computing in Python. It provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. NumPy is essential for data analysis, machine learning, and numerical computations.

7) PIP

PIP is the package installer for Python. It allows users to install and manage additional libraries and dependencies that are not included in the standard Python library. PIP simplifies the process of downloading and installing packages from the Python Package Index (PyPI).

III. BLOCK DIAGRAM

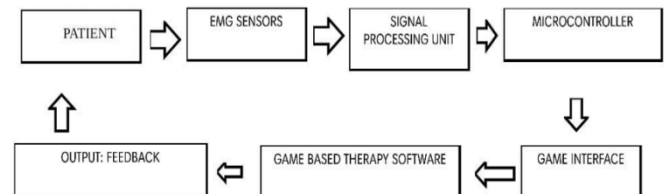


Fig 1: Block diagram

IV. COMPARISON OF MUSCLE REFLEXES OF HEALTHY PERSON AND CEREBRAL PALSY PATIENT

Muscle reflexes in healthy individuals and those with cerebral palsy exhibit significant differences due to variations in motor control and neurological function. In a healthy person, reflexes are well-coordinated and balanced, responding appropriately to external stimuli through controlled muscle contractions and relaxation. These reflexes are essential for maintaining posture, balanced and smooth motor control

In contrast, individuals with cerebral palsy often experience disturbances in their muscle reflexes due to motor impairment caused by brain damage or abnormal

development. This condition can lead to abnormal reflex responses, including hyperactive reflexes that cause muscle stiffness, spasticity, and involuntary movements. In some cases, certain reflexes may be diminished or absent, making it difficult to control and coordinate voluntary movements.

The differences in muscle reflexes between healthy individuals and those with cerebral palsy significantly impact motor function and daily activities. While healthy reflexes enable fluid and precise movement, individuals with cerebral palsy may struggle with executing controlled motions. The severity of these reflex abnormalities varies depending on the type and extent of cerebral palsy. As a result, therapeutic interventions focus on managing these impairments by improving muscle tone, reducing spasticity, and enhancing overall motor function. Effective rehabilitation strategies, including physical therapy, medication, and assistive technologies, play a crucial role in improving mobility and independence for individuals with cerebral palsy.

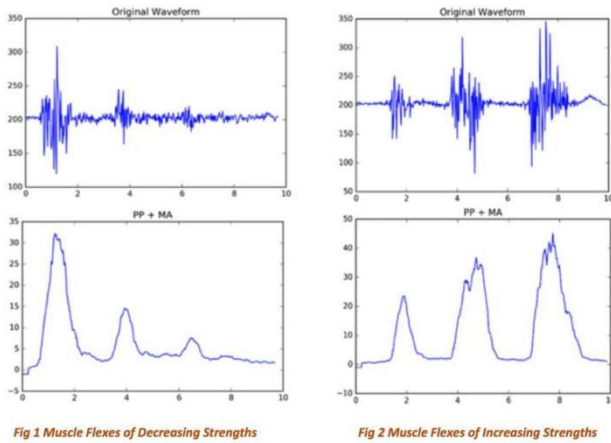


Fig 2: Muscle Flexes Of Increasing And Decreasing Strength

V. FACE RECOGNITION APPLICATION

Face recognition technology is widely used across various industries for security, convenience, and efficiency. In security, it enables access control by allowing authorized individuals to unlock devices, secure areas, and buildings. Law enforcement agencies utilize it for identifying and tracking suspects through surveillance footage, aiding investigations.

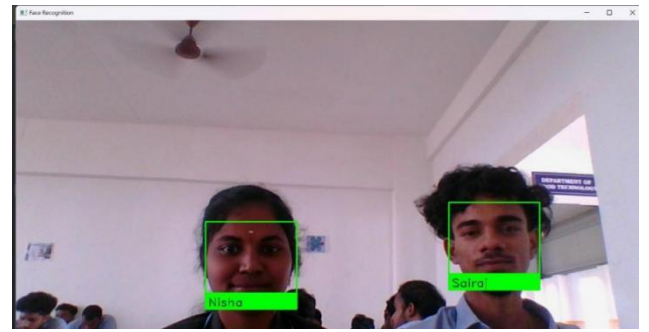


Fig 3: Face Recognition (Sign In)

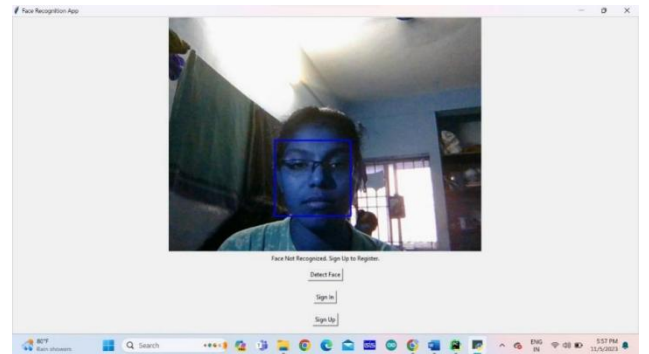


Fig 4: Face Recognition (New Patient)

VI. HAND RECOGNITION

Hand recognition technology plays a crucial role in rehabilitation, particularly for individuals recovering from neurological conditions such as stroke and cerebral palsy. By tracking hand movements and gestures, this technology enables interactive therapy sessions that improve motor skills, coordination, and muscle strength.

In home-based rehabilitation, hand recognition can be integrated with game-based therapy, allowing patients to engage in exercises tailored to their specific needs. It provides real-time feedback, helping users refine their movements and track progress over time. This approach enhances patient engagement, ensures accurate therapy execution, and supports a more effective recovery process.

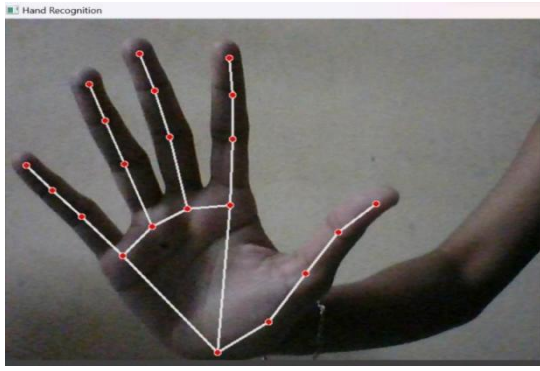


Fig 5: Hand Recognition

VII. CONCLUSION

EMG-controlled game-based therapy is redefining rehabilitation by making recovery interactive, engaging, and accessible. By combining real-time feedback with personalized exercises, this approach enhances motivation, improves motor function, and enables patients to continue therapy from home without frequent hospital visits.

To further advance this innovation, a Windows application powered by Node.js and TensorFlow will integrate EMG-controlled gameplay, offering adaptive difficulty and precise progress tracking. This technology-driven solution not only accelerates recovery for stroke and cerebral palsy patients but also paves the way for a future where rehabilitation is smarter, more effective, and truly patient-centric.

VIII. EMG-CONTROLLED GAME

The game interface integrates Electromyography (EMG) technology to provide an interactive and engaging rehabilitation experience. The system detects muscle activity through EMG sensors, allowing users to control in-game actions based on their muscle movements. This not only makes therapy enjoyable but also actively engages patients in their rehabilitation process.

As seen in the interface, the game presents a simple yet effective design, where users perform controlled muscle movements to interact with game elements. In this case, the goal is to catch falling objects (such as eggs) in a basket, requiring precise and coordinated muscle contractions. This setup helps improve motor control, reflexes, and hand-eye coordination—key areas affected in conditions like stroke and cerebral palsy.

By integrating real-time feedback and adaptive difficulty, the EMG-controlled game ensures that therapy is both challenging and rewarding. This innovative approach transforms rehabilitation into an engaging experience, promoting motivation, consistency, and ultimately, faster recovery.

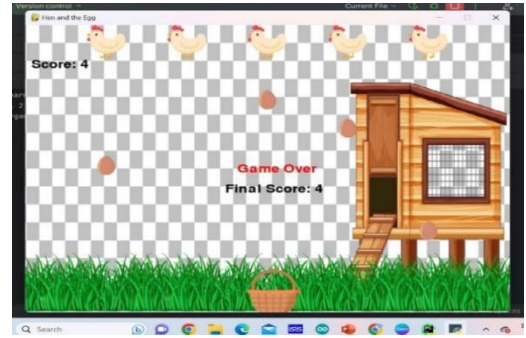
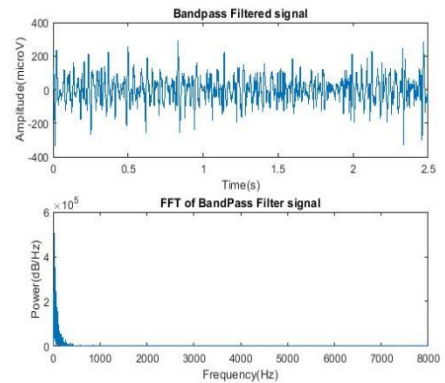
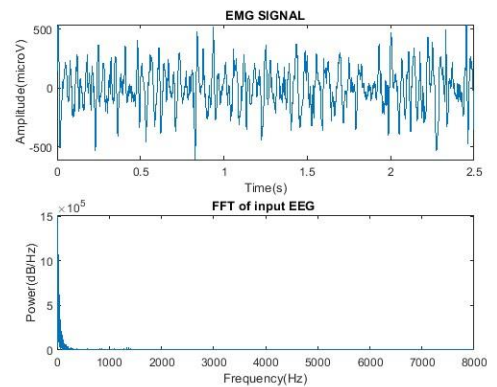


Fig 6: Egg Catcher (Game Based Therapy)

IX. OUTPUT



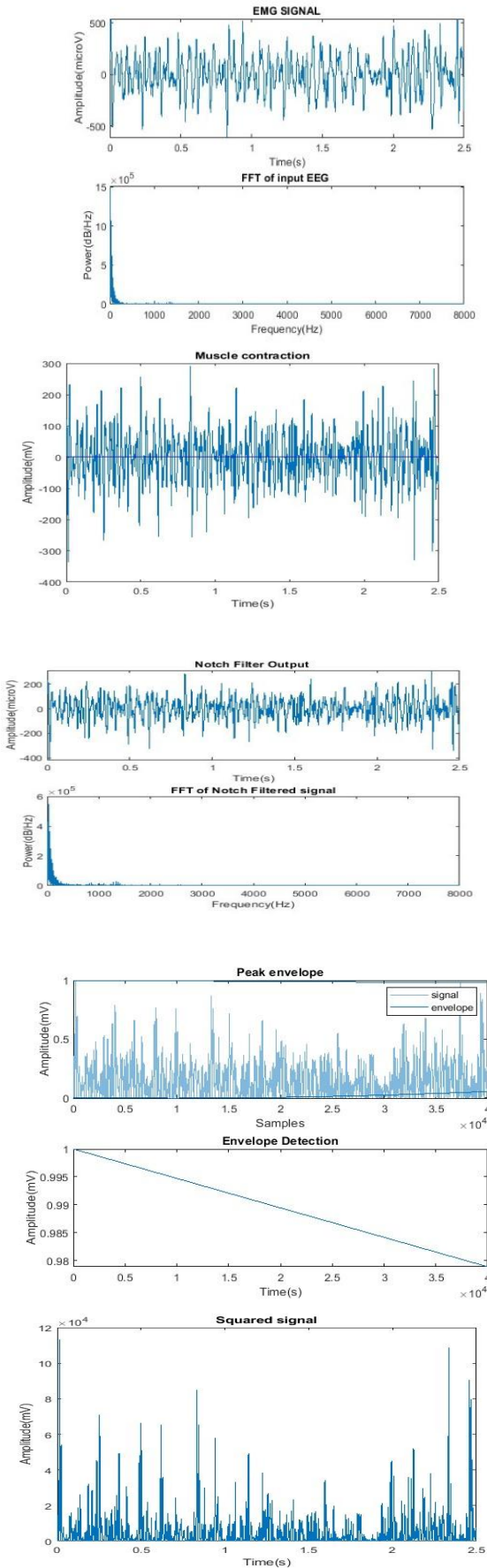


Fig 7: Various EMG Signals With Notch Filter, Band Pass , Peak EMG Etc..

X. FUTURE WORKS

We aim to develop a complete prototype integrating EMG-controlled game-based therapy for stroke and cerebral palsy rehabilitation at home. This includes refining the user interface, optimizing sensor accuracy, and designing adaptive games for different motor impairments.

To enhance accessibility, an Android app will be introduced, enabling therapy anytime, anywhere. Real-time progress tracking and AI-driven feedback will help personalize therapy and assist healthcare providers in monitoring patient recovery. Future improvements will focus on minimizing sensor latency, enhancing game responsiveness, and exploring multiplayer rehabilitation modes for remote engagement.

These advancements will make home-based rehabilitation more effective, engaging, and personalized for better recovery outcomes.

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